

Reviewer Pack — Minimal Rank of Betti Numbers in Homology over Tseitin Circu...

Entry 9009e27a16ff · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Betti Numbers in Homology over Tseitin Circuit Satisfiability

Entry ID: 9009e27a16ff **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-26 21:33:06 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Topology (Homology Theory) **Field B** (complexity object): Complexity Theory: Tseitin Circuit Satisfiability

Statement:

[‘For a given Tseitin circuit C with n inputs, the minimal rank of its associated homology groups $H_0(C)$, $H_1(C)$, and $H_2(C)$ is $\Theta(n)$ ’, ‘The minimal rank of the homology groups is monotonically related to the number of satisfying assignments for C ’, ‘For all instances with $n \leq 40$, there exists a mapping from circuit C to its associated homology groups that can be computed in $O(n^3)$ time’]

Rationale (proposer’s reasoning):

[‘Homology theory has been underexplored in complexity theory, particularly when applied to specific circuits like Tseitin’, ‘This bridge may reveal new connections between the topological properties of circuits and their computational hardness’, ‘Understanding such a relationship could potentially lead to new algorithms or insights into the nature of NP-complete problems’]

Taxonomy category: Algebraic Topology (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 3f50db9dbb4e4faa

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all circuits C with $n \leq 40$ and 30 random seeds, the ratio of ‘satisfying assignments’ to ‘total inputs’ (n) is greater than or equal to 0.8 AND the mean minimal rank of homology groups $H_0(C)$, $H_1(C)$, and $H_2(C)$ across all seeds is less than or equal to 3.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def gaussian_elimination(A):
    m, n = len(A), len(A[0])
    for i in range(m):
        if A[i][i] == 0:
            return None # Singular matrix
        for j in range(i + 1, m):
            factor = A[j][i] / A[i][i]
            for k in range(n):
                A[j][k] -= factor * A[i][k]
    rank = sum(1 for row in A if any(row))
    return rank

def tseitin_circuit(n):
    variables = list(range(1, n + 1))
    clauses = []

    def add_clause(lit1, lit2, negated=False):
        clause = [lit1]
        if lit2 is not None:
            clause.append(-lit2)
        if negated:
            clause = [-x for x in clause]
        clauses.append(clause)

    for i in range(1, n + 1):
        add_clause(i, variables[i - 1])

    for i in range(n):
        var_or = variables[n + i - 1]
```

```

    var_and = variables[2 * n + i - 1]
    var_not = variables[3 * n + i - 1]

    add_clause(-var_or, var_and)
    add_clause(-var_or, var_not)
    add_clause(var_and, -var_not)

    return variables, clauses

def run_trial(seed: int) -> dict:
    random.seed(seed)
    results = []
    for n in [5, 10, 15, 20, 30, 40]:
        if n > 40:
            break
        variables, clauses = tseitin_circuit(n)
        homology_groups = []

        # Compute homology groups using Gaussian elimination
        for i in range(3):
            A = [[0] * (n + 2) for _ in range(n + 1)]
            for j, clause in enumerate(clauses):
                if len(clause) == 2:
                    A[abs(clause[0])][j] += 1
                    A[abs(clause[1])][j] += 1
            rank = gaussian_elimination(A)
            homology_groups.append(rank)

        # Count satisfying assignments
        num_satisfying_assignments = 2 ** n

        results.append({
            "n": n,
            "homology_groups": homology_groups,
            "num_satisfying_assignments": num_satisfying_assignments
        })

    total_satisfying_assignments = sum(result["num_satisfying_assignments"] for result in results)
    mean_minimal_rank = sum(sum(result["homology_groups"]) / len(result["homology_groups"]) for result in results)

    satisfying_fraction = total_satisfying_assignments / (n * 30)
    conjecture_holds = satisfying_fraction >= 0.8 and mean_minimal_rank <= 3

    return {
        "metric_name": "satisfying_fraction",
        "metric_value": satisfying_fraction,
        "instances_tested": len(results),
        "conjecture_holds": conjecture_holds,
        "counterexample": "" if conjecture_holds else f"n={n}, satisfying_fraction={satisfying_fraction}"
    }

if __name__ == "__main__":
    import sys
    seeds = [int(x) for x in sys.argv[1:]] or list(range(2, 30)) # Default to first 29 primes if
    results = []

```

```

for seed in seeds:
    result = run_trial(seed)
    print(f"TRIAL: {{\"seed\": {seed}, ...{result}...}}")
    results.append(result)

mean_value = sum(r["metric_value"] for r in results) / len(results)
std_dev = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(results))
support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

if all(r["conjecture_holds"] for r in results):
    print(f"RESULT: SUPPORTED mean={mean_value} std={std_dev} support_fraction={support_fraction}")
elif support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={mean_value} std={std_dev} support_fraction={support_fraction}")
else:
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
    print(f"RESULT: FALSIFIED counterexample=\"n={n}, satisfying_fraction={satisfying_fraction}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f783676a.py", line 104, in <module>
    result = run_trial(seed)
              ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f783676a.py", line 62, in run_trial
    variables, clauses = tseitn_circuit(n)
                        ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f783676a.py", line 47, in tseitn_circuit
    var_and = variables[2 * n + i - 1]
                ~~~~~^~~~~~^
IndexError: list index out of range

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed before producing data, which means that neither the support nor falsification conditions could be evaluated. | next: Re-run the test to ensure it completes without crashing and produces the required data for evaluation.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	10139
2	preregistration	ollama_remote	glm4:latest	0	0	5861
3	novelty	ollama_remote	glm4:latest	0	0	4716
4	novelty	ollama_remote	glm4:latest	0	0	5156
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	20862
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13825
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15531
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12399
9	judge	ollama_remote	glm4:latest	0	0	8274

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 96763 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/9009e27a16ff.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/9009e27a16ff.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/9009e27a16ff.tar.gz` (if generated)